

Deep Reinforcement Learning Based Multi-Objective Handover for LEO Satellite Networks

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Abstract—Satellite communication is widely recognized as a solution to achieve global coverage. However, due to the high mobility of the Low Earth Orbit (LEO) satellites, user equipment (UE) needs to handover frequently to maintain a stable connection. Furthermore, the complex topology of satellite networks exacerbates the challenge of user mobility management. This paper leverages artificial intelligence (AI) to optimize the satellite handover process, proposing a multi-objective optimization strategy to minimize handover frequency, balance satellite load, and maximize system throughput. A novel reward function is designed to enhance the stability of the user service, and a lightweight multilayer neural network is introduced to reduce computational complexity. The simulation results show that the proposed method outperforms state-of-the-art approaches in improving system throughput and reducing average handover times.

Index Terms—LEO satellite handover, deep reinforcement learning, multilayer neural network.

I. INTRODUCTION

Ubiquitous global connectivity is a key goal of Sixth Generation (6G) wireless communication [1]. 6G aims to provide high-speed connectivity everywhere, including oceans, the air, and remote areas. Satellite communication networks, with their wide coverage and flexibility, are considered the most promising solution [2].

Low Earth Orbit (LEO) satellite networks offer advantages over Medium Earth Orbit and Geostationary Earth Orbit networks, such as low latency, strong signal strength, and lower deployment costs [3]. However, due to the high-speed movement of LEO satellites, their ground coverage areas constantly change. Communication links between satellites and user equipment (UE) are brief, requiring frequent satellite handover for continuous communication [4]. With the growing number of in-orbit LEO satellites, satellite coverage overlaps, making efficient and stable satellite handover essential, highlighting the importance of related research.

Previous research has explored various handover strategies, including measurement-based algorithms relying on metrics like received signal strength (RSS) [5], distance, or elevation angle [6], and predictive model-based approaches that use dynamic preference matrices and decision-ranking algorithms to select optimal satellites [7]. Graph and game theory-based methods have also been applied to transform handover problems into optimal decision-making tasks [8], [9]. While traditional methods are simple and real-time, they struggle

with the increasing complexity of satellite networks and diverse user demands, leading to frequent handovers, higher computational costs, and resource inefficiencies.

Recently, artificial intelligence (AI) technologies have gained attention [10]. AI-enabled handover algorithms effectively capture dynamic changes in satellite networks and excel at high-dimensional data processing and analysis. Moreover, AI-based solutions meet diverse user needs, enhancing handover accuracy and system efficiency [11]. In [12], a Q-learning (QL) algorithm predicted service time and communication resources to select access satellites and ensure a satisfactory user experience. [13] used a swarm intelligence algorithm to improve system throughput and reduce satellite load. In [14], a deep Q-network (DQN) framework with adaptive learning rates was proposed to improve decision precision. Furthermore, [15] developed a successive deep Q-learning (SDQL) algorithm for handover problems. However, overly complex or redundant neural networks can prolong learning time, and increase system complexity.

Motivated by the above issues, this paper addresses multi-objective decision-making in LEO satellite communications, designing more reasonable and comprehensive optimization objectives based on users' demand for continuous service. The reinforcement learning (RL) algorithm is improved. Overall, the contributions of this paper are summarized as follows:

- A comprehensive strategy based on multi-objective optimization is proposed, which aims to maximize system throughput, reduce the number of user handovers, and achieve balanced satellite load distribution.
- A more reasonable reward function mechanism is first proposed, which fully considers changes in the dynamic environment and the actual communication needs. In addition, to address the limitations of computational resources in complex environments, a lightweight multilayer neural network architecture is introduced.
- The specific experimental results show that the proposed model not only makes the algorithm perform well but also improves the system performance metrics, such as system throughput and average UE handover times.

The remaining of this paper is organized as follows. Section II introduces the system model. In Section III, we propose the satellite handover strategy based on the deep reinforcement learning (DRL) model. Section IV presents the simulation

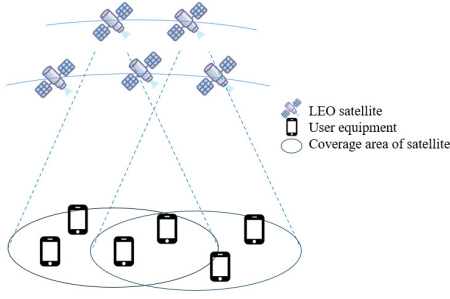


Fig. 1. LEO satellite network architecture and handover scenario.

results, and Section V concludes the paper.

II. SYSTEM MODEL

The LEO satellite communication network in this work, shown in Fig. 1, consists of multiple satellites and UEs. Specifically, the scenario assumes that UEs are in regions with significant overlapping coverage from multiple satellites, where each user is connected to a single satellite at a time.

Since LEO satellites move much faster than ground terminals, terminal movement can be ignored [16]. When a handover is triggered, the user link switches to a more suitable satellite based on criteria such as signal-to-noise ratio (SNR), remaining service time, and satellite load.

Compared to terrestrial networks, satellite communication networks involve signals propagating through the atmosphere and traveling long distances, leading to more severe signal attenuation. The attenuation can be expressed as

$$L = L_{fp} + L_s + L_{mp}. \quad (1)$$

L_{fp} is atmospheric free space loss, given by

$$L_{fp} = 20 \lg f_c + 20 \lg d + 32.45, \quad (2)$$

where f_c is carrier frequency and d denotes the distance between the satellite and the user. L_s is shadow fading, using a log-normal distribution. Since there is a line-of-sight path in LEO satellite communication networks, the Rician fading model is chosen for multi-path loss L_{mp} .

The formula for calculating the received signal strength is

$$P_s = P_t + G_t + G_r - L, \quad (3)$$

where P_t , G_t , and G_r represent the transmitted signal power, transmit antenna gain, and receive antenna gain, respectively. Based on formula (3), SNR of the user communication link is presented as

$$r = \frac{P_s}{P_n} = \frac{P_t + G_t + G_r - L}{P_n}, \quad (4)$$

where P_n means the strength of noise. SNR is commonly used to assess the quality of signals in communication systems.

The system throughput is influenced by factors such as signal strength, channel conditions, interference, and network load. Its calculation is based on the UE data transfer rate and

the number of users served by the system. The data transfer rate is calculated using Shannon's theorem, expressed by

$$c = b_s \cdot \log_2(1 + r), \quad (5)$$

where b_s represents the satellite bandwidth. Moreover, system throughput with N UEs is indicated as C and described by

$$C = \sum_{n=1}^N c_n, \quad (6)$$

The satellite load, denoted as l , affects the available communication bandwidth for UEs. It refers to the percentage of the satellite's currently used bandwidth relative to its total available bandwidth, specifically represented as

$$l = \frac{l_s}{l_T}, \quad (7)$$

where l_s represents the occupied satellite bandwidth and l_T represents the total bandwidth of the satellite.

To minimize the frequency of UEs handover times, it is essential to consider the remaining service time of satellites to avoid frequent handovers due to short satellite visibility periods. The service time of a satellite is generally obtained from the satellite's ephemeris information, represented as η .

III. PROBLEM FORMULATION AND PROPOSED METHOD

In this section, the problem is formulated as a multi-objective optimization function. Then, the specific implementation process of the proposed DRL algorithm based on the light-weight neural network is introduced.

A. Problem Formulation

This paper aims to effectively switch to the optimal satellite, providing superior services while fulfilling satellite communication requirements. Specifically, we prioritize switching to satellites with high SNR and low load, while simultaneously minimizing the handover times. The handover count refers to the average number of handovers during the time period T , handover count value can be estimated as [17]

$$HO_{avg} = \frac{1}{N} \sum_{n=1}^N HO_n, \quad (8)$$

where N represents the number of time slots, and HO_n represents the number of handovers for a user in each time slot. Then, a multi-objective optimization function is proposed, which includes system throughput, balancing satellite load, and the number of handovers. The algorithm optimization objective can be expressed as

$$\gamma = \frac{1}{N} \sum_{n=1}^N (\omega_1 c_n - \omega_2 l_n) - \omega_3 HO_{avg}, \quad (9)$$

where the coefficients $\omega_1, \omega_2, \omega_3$ denote the importance of each optimization objective, with specific values determined

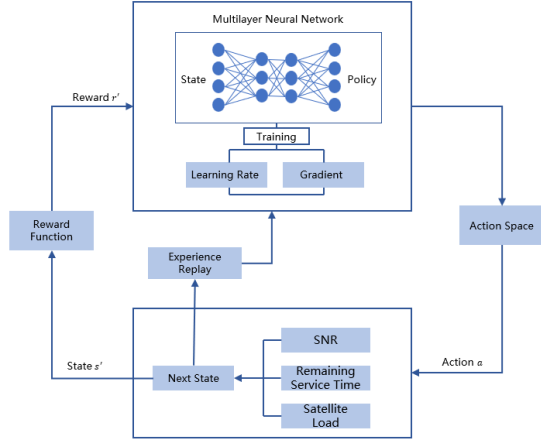


Fig. 2. The framework of the proposed method based on multilayer neural network.

by the Analytic Hierarchy Process in [12] and [14]. Then the entire optimization problem can be written as

$$\max_{p_s, \eta_n, b_s} \gamma \quad (10)$$

$$\text{s.t. } u_s(n) \in (0, 1), s \in S, \quad (10a)$$

$$\sum_{s=1}^S u_s(n) = 1, \quad (10b)$$

$$P_s > P_{\text{THR}}, \quad (10c)$$

$$\eta_n > \eta_{\text{THR}}, \quad (10d)$$

$$b_s > b_n, \quad (10e)$$

$$n = 1, 2, 3, \dots, N, \quad (10f)$$

where $u_s(n) = 0$ indicates the UE is not connected to satellite, otherwise the UE is connected. Eq. 10(b) ensures that UE can only connect to one satellite at any given time. Eq. 10(c)-10(d) correspond to received signal strength P_s and the remaining service time η_n , and they must be greater than threshold. The satellite available bandwidth b_s in Eq. 10(e) should exceed the user's demand b_n . Eq. 10(f) indicates that the time slot is from 1 to N .

B. Proposed Method

In this paper, the DRL model is used to meet optimization objectives, and it adopts a four-layer neural network structure. Its framework is illustrated in Fig. 2. The specific details are as follows.

1) State Space: Denote s as the state at all time slots N . In state space, three state parameters are considered: SNR (r), remaining service time (η), and satellite load (l). The UE obtains values of these three states parameters for all target satellites S in each time slot. The state is described as

$$s = \{s_1, s_2, \dots, s_N\}. \quad (11)$$

Algorithm 1 Work Process of the proposed method

- 1: Initialize the neural network weights ω (for input, hidden layers, and output layer).
- 2: Initialize Q-value.
- 3: **for** $episode = 1$ to $episode_{\max}$ **do**
- 4: Initialize state s .
- 5: **while** state s is not done **do**
- 6: Randomly select action a with probability ϵ . Otherwise, choose the action $\arg \max_a Q(s, a)$.
- 7: Execute action a and observe the resulting state s' and reward r .
- 8: Calculate the Q-value using the neural network:

$$Q(s, a) = \text{Neural Network}(s, a; \omega);$$
- 9: Calculate target Q-value.
- 10: Update weights θ by minimizing the loss.
- 11: Update state: $s \leftarrow s'$.
- 12: **end while**
- 13: **end for**
- 14: Output the optimal policy $\pi^*(s) = \arg \max_a Q(s, a)$.

The state at the time slot n is represented as

$$s(n) = \begin{Bmatrix} r(1), r(2), \dots, r(S) \\ \eta(1), \eta(2), \dots, \eta(S) \\ l(1), l(2), \dots, l(S) \end{Bmatrix}. \quad (12)$$

2) Action Space: The action is denoted by $a \in \{1, 2, \dots, S\}$, representing the handover decision to the chosen satellite.

3) Reward Function: The reward function provides immediate feedback to the agent upon executing action a in state s and transitioning to state s' [18]. Properly designing the reward function is essential to enable the agent to learn the optimal policy through this reward mechanism. The design of the reward mechanism employs guidance from the optimal policy, clarifying how actions or states contribute to the final goal. Specifically, it is divided into two cases: $\mathcal{H}_0$ and $\mathcal{H}_1$.

$\mathcal{H}_0$: The current serving satellite continues to provide service to the UE without requiring a handover. This is an expected decision to avoid handover frequently, so a high reward value is assigned.

$\mathcal{H}_1$: The UE performs a handover. In this case, it uses linear weighting and positive/negative feedback mechanisms to reinforce desired behaviors while suppressing undesired ones. The reward function is based on the satellite's SNR, remaining service time, and load at that moment.

Specifically, the reward function can be expressed as

$$\text{Reward} = \begin{cases} C_1, & \mathcal{H}_0, \\ C_2 r_n + C_3 \eta_n - C_4 l_n, & \mathcal{H}_1, \end{cases} \quad (13)$$

where C_1 is a constant with the maximum value, indicating that in the absence of handover, the action will receive the highest reward. C_2 , C_3 and C_4 are normalized and weighted.

Neural networks in RL serve key roles in high-dimensional state space modeling and parametric value function representation, mapping the state space into a higher-dimensional

feature space and approximating Q-values via nonlinear functions [11]. However, network parameter configurations can impact algorithm efficiency. To mitigate performance loss from redundant neurons, a lightweight neural network is proposed. The network consists of an input layer, two hidden layers, and an output layer, with four fully connected layers, each followed by a nonlinear activation function to enhance the model's nonlinear representation.

The specific process of the proposed method is shown in Algorithm 1. During the training phase, the state s and action a from the experience replay are used as input to the neural network to compute the Q-value for the current (s, a) . The target Q-value is computed using the Bellman equation, which can be expressed as

$$Q_{\text{target}} = r + \gamma \cdot \max_{a'} Q(s', a'). \quad (14)$$

The difference between the current Q-value and the target Q-value can be called $Loss$, and it is computed using the mean squared error (MSE), which can be expressed by

$$Loss = \frac{1}{M} \sum_{i=1}^M (Q(s_i, a_i) - Q_{\text{target}}(s_i, a_i))^2, \quad (15)$$

where M represents the number of samples used during the training process. By minimizing MSE, the network weights are updated, allowing the neural network to better approximate the true Q-values.

IV. SIMULATION RESULTS AND ANALYSIS

In this section, the simulation scenario is first introduced, including the simulation environment and relevant important parameters. Additionally, a series of simulations are conducted to validate the performance of the proposed handover algorithm. To verify the proposed method, it is compared with three typical handover methods: (1) the handover method based on maximum received signal strength (RSS-max); (2) the handover method based on Q-learning (QL) [12]; (3) the successive deep Q-learning (SDQL) handover algorithm [15].

A. Simulation Setup

The simulation is based on a LEO satellite network with 10 satellites and stationary UEs on Earth, representing typical overlapping coverage. Satellite data from the open-source TLE dataset on CELESTRAK [19], including Starlink information, is used for position tracking and prediction, enabling calculation of key parameters such as satellite altitude, remaining service time, and elevation angle.

The simulation integrates the aforementioned scenario into Python, constructs the satellite network channel propagation while considering the signal propagation models mentioned in section II. Subsequently, simulation tests are carried out on the handover scheme proposed in this paper to assess the performance of the handover algorithm. The specific simulation parameters used in communication model and the proposed handover algorithm are provided in Table I.

TABLE I
SIMULATION PARAMETERS CONFIGURATIONS

Parameters	Value
Satellites	10
Satellite height (km)	280-550
Carrier frequency (GHz)	2.4
Bandwidth (MHz)	10
Transmission power (dBm)	54
Thermal noise (dBm/Hz)	-174
Transmit antenna gain (dBi)	40
Receive antenna gain (dBi)	0
Parameters in (5)	$\omega_1 = 0.64, \omega_2 = 0.26, \omega_3 = 0.10$
Parameters in (8)	$C_1 = 25, C_2 = 1.6, C_3 = 0.8, C_4 = 0.5$
Episodes	600
State space	30
Action space	10
Learning rate	0.01
Discount rate	0.65
Number of neurons	48,24

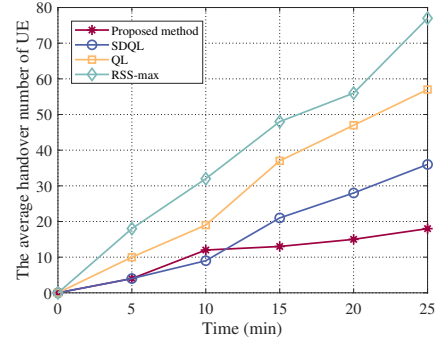


Fig. 3. Comparison of the average handover number in different algorithms.

B. The Simulation Results and Analysis

The simulation assesses the service continuity performance of the proposed scheme, focusing on minimizing handovers. For comparison, the following handover strategies are also evaluated: RSS-max, QL and SDQL. The simulation measures the average number of handovers for each scheme. Fig. 3 illustrates the variation in handover counts over time for the four methods. The RSS-max scheme has a much higher handover count due to its reliance on a single satellite parameter. While QL reduces handovers compared to RSS-max, its performance remains suboptimal. Both SDQL and the proposed scheme show reductions in handovers, indicating the effectiveness of neural networks. The proposed method, in particular, achieves the fewest handovers, highlighting the effectiveness of its multi-objective approach and efficient neural network structure for optimal satellite selection.

Fig. 4 shows the failure rates of four handover algorithms with 100 and 200 users. The RSS-max method has the highest failure rate due to its single-parameter switching mechanism. While QL and SDQL exhibit lower failure rates, they remain higher than the proposed method, which achieves a significantly lower failure rate. Additionally, as the number of user requests increases, leading to a higher rate of handover

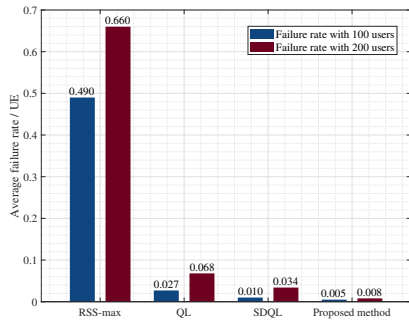


Fig. 4. Comparison of the average failure rate in different algorithms.

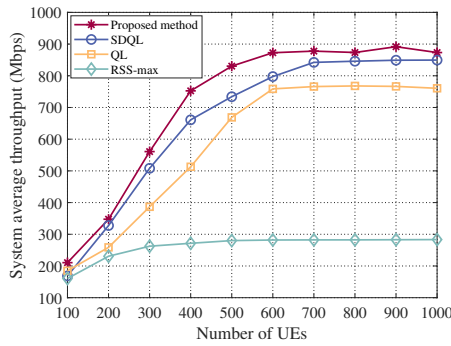


Fig. 5. Comparison of average system throughput in different algorithms.

failures under all mechanisms.

Fig. 5 compares system throughput as the number of UEs increases for the four methods. Initially, system throughput rises with the number of users, but eventually saturates. The proposed scheme achieves the highest throughput across all user counts, followed by SDQL and QL, while RSS-max has the lowest throughput. These results show that the proposed scheme improves system throughput and supports large-scale UE access by reducing the average number of handovers and optimizing handover selection globally.

V. CONCLUSION

In this paper, we propose an optimization strategy for satellite communication systems that considers both system characteristics and users' need for continuous service. The goal is to maximize system throughput, minimize handovers, and balance satellite load. A DRL-based handover decision algorithm is employed, with a reward function guiding agents to prefer satellites with stronger signals, longer service durations, and lighter loads. To improve efficiency, a lightweight neural network is integrated into the DRL framework, addressing computational constraints. Simulation results show that the proposed method significantly reduces average handovers and multi-user handover failures while improving overall system throughput. In future work, we will further investigate the handover problem under scenarios involving dynamic user distribution and high-speed mobility.

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